NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2024-2025

SC2006/CZ2006 – SOFTWARE ENGINEERING

Nov/Dec 2024 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 4 pages.
2. Answer **ALL** questions.
3. This is an open-book examination.
4. All questions carry equal marks.
5. Refer to Appendix A on page 4 for the project description which is needed to answer some of the questions.

---

1. Based on the project description given in Appendix A,

   (a) Identify the actors and use cases of the project and draw the use case diagram. Use <<include>> and <<extend>> relationships where appropriate.

   (19 marks)

   (b) Write the use case description for the functionality of the task assignment. Your use case description must include the following elements: Actors, Pre-conditions, Flow of events and Alternative flows, if any.

   (6 marks)

2. (a) For the functionality of the task assignment, identify the main classes and their associations and draw a conceptual class diagram. Your conceptual class diagram should clearly depict the stereotypes of each class. You do not need to identify any attributes or operations within the classes.

   (9 marks)

Note: Question No. 2 continues on Page 2

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

(b) Draw an activity diagram with swim lanes that shows the activities and transitions of the system from when a task is created to when it is deleted.

(7 marks)

(c) Suppose a team is following a Waterfall approach but has realized halfway through the project that the requirements have drastically changed. What steps should the team take to address these changes?

(4 marks)

(d) You are a project manager tasked with developing a software application using an Agile process. The team has just completed the first sprint, but the customer is unhappy with the progress. How would you handle this situation to ensure that customer satisfaction improves in future sprints?

(5 marks)

3. (a) Based on the project description given in Appendix A,

(i) Propose an appropriate architecture for the Task Management System, and draw a detailed Class diagram with key attributes and methods in each class to reflect the architecture design.

(8 marks)

(ii) Identify the one relevant non-functional requirement for the system. Propose an alternative architecture design to address this non-functional requirement.

(8 marks)

(b) Answer the following questions related to software design:

(i) Propose a scenario in the Task Management System where more than one design patterns can be applied together, and draw the class diagram to explain the scenario.

(6 marks)

(ii) Design patterns can be combined together sometimes. Is it always good to combine different design patterns? Explain your answer.

(3 marks)

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

4. (a) When a new task is created in the Task Management System, the following information is required:

1. Task Description: the detailed task description, which must be in ASCII format.
2. Resource needed: a 4-digit integer value.
3. Deadline: a date between 2024 and 2026.
4. Task Priority: one of "Critical", "High", "Normal", "Low" and "Very Low".

(i) Determine the equivalence classes for the above FOUR inputs.

(4 marks)

(ii) Determine the boundaries of the equivalence classes identified in your answers to Q4(a)(i). For each boundary, identify a value on the boundary, a value just below the boundary, and a value just above the boundary.

(4 marks)

(iii) You intend to perform **defensive testing** of the task information. Design a set of test cases to test the FOUR inputs based on the equivalence classes and boundary values identified in your answers to Q4(a)(i) and Q4(a)(ii).

(5 marks)

(b) Shift left is a methodology in software development that emphasizes moving testing activities earlier in the development cycle than in a traditional development approach where testing happens later right before deployment. Explain why this can be a good methodology.

(4 marks)

(c) Software maintenance is often the most resource-intensive phase in the software lifecycle. Elaborate on four challenges that may arise during the maintenance phase and suggest best practices to mitigate these challenges.

(8 marks)

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

Appendix A

Web-Based Task Management System for Collaborative Teams

The proposed project aims to develop a user-friendly, web-based task management system designed to enhance collaboration, productivity, and communication for teams. This system will allow users to create, assign, and monitor tasks in real-time while promoting transparency and accountability within a team or organization. The primary focus of the system is to streamline task coordination, optimize resource allocation, and support agile methodologies like Scrum, making it particularly beneficial for teams working on software development, marketing campaigns, product launches, and more.

To use the system, all the team members should be registered to the system because they need to login to the system before they can use the system. Role-based access control will ensure that users have appropriate levels of access to the system, safeguarding sensitive information.

The system shall allow team members to create tasks. Team leaders or project managers will verify them and assign them to specific individuals or teams and set deadlines or milestones. Tasks can be categorized, labeled with priorities, and tagged for easy retrieval. A task can be updated and deleted after it is finished. A task can also be re-assigned to other individuals or teams. But a task cannot have two assignments at the same time.

Team members shall collaborate on tasks via built-in chat, comment threads, and document sharing, ensuring that all relevant stakeholders are informed of progress.

An intuitive dashboard will display task progress, overdue tasks, and workload distribution, providing team leaders and project managers with clear visibility into team performance.

The system shall notify team members about deadlines, upcoming tasks, and priority changes via an email server and in-app notifications. Customizable alerts can be set to remind users of important tasks or milestones.

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

END OF PAPER

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library

CZ2006 SOFTWARE ENGINEERING
SC2006 SOFTWARE ENGINEERING

CONFIDENTIAL

Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

ATTENTION: The Singapore Copyright Act applies to the use of this document. Nanyang Technological University Library